Ampoule

ZERO-KNOWLEDGE PRESCRIPTIONS ON MIDNIGHT

One prescription, one fill, *no patient data on chain.*

Monthly caps on opioids and other controlled drugs that hold across every doctor and every pharmacy, enforced by a Compact contract that never learns who takes what.

THE PROBLEM

A controlled-drug prescription is *two things at once*

A medical record

Tramadol after a knee injury, alprazolam for anxiety, methylphenidate for ADHD. It should stay between the patient, the doctor and the pharmacist.

A claim on a supply that gets abused

The same prescription can be copied, edited, filled twice, or collected again from a second doctor and a second pharmacy. Society wants that policed.

Today a patient gets one or the other: rules that are easy to beat, or rules enforced by a list of everyone on painkillers.

TODAY

Both of today's answers fail

Paper and PDF prescriptions

Each pharmacy sees one slip and nothing else, so doctor shopping is invisible to all of them.

0.7%

of US opioid buyers obtained 1.9% of all opioid prescriptions, from 10 prescribers each on average.

A named register

Every Indian pharmacy keeps a Schedule H1 register: the patient's name, the drug and the quantity for each listed sale, kept for three years. No other pharmacy can see it.

31

US states let police query the prescription monitoring database without a warrant.

McDonald and Carlson, PLOS ONE 2013 (2008 data). Drugs and Cosmetics Rules, 1945, Schedule H1 (2013). ACLU on prescription drug monitoring programs. Full list in docs/ADOPTION.md.

AMPOULE

The rules, enforced by the chain. *The register, never built.*

One prescription, one fill

A copied or screenshotted prescription is refused everywhere: `ALREADY_FILLED`.

A cap per person, network-wide

One opioid fill per 30 days from any doctor at any pharmacy: `CAP_REACHED`.

Licences that bite

Suspend a doctor and every open prescription they wrote stops at once: `DOCTOR_SUSPENDED`.

The chain never sees who the patient is, the drug, the quantity, the expiry date, or anything that links two fills by one person. Refused attempts never reach it at all.

HOW IT WORKS

A prescription's life, in five steps

DOCTOR

Issues

Proves the key the medical council registered for this licence, and appends one hash of the prescription to a Merkle tree.

PATIENT'S PHONE

Holds it

Receives the prescription as a QR code, like the paper slip. It is bound to the phone's patient key.

PHARMACY

Asks

Shows a QR with its licence number and a one-time challenge.

MIDNIGHT

Fills

The phone proves every rule in one transaction. Two nullifiers and a sealed record are written. Nothing else.

PHARMACIST

Checks

The phone opens the sealed record for the pharmacist alone: exactly this drug, this quantity. Hand it over.

The doctor sees what they prescribe and the pharmacist sees what they hand over. The ledger sees licence numbers, a drug class and hashes.

THE CORE CIRCUIT

One proof at the counter

The `fill` circuit proves, privately

- the patient key is enrolled (patient tree)
- the prescription was issued (prescription tree) to this key
- its doctor is licensed and active; the pharmacy too
- it has not expired, and this fill number is unused
- this person is under the cap for this class in this window

```
// spent once, per fill of a prescription
fill = H(patientSecret, rx, k)
// one per slot, per class, per 30-day window
cap  = H(patientSecret, class, window, slot)

// published, each through disclose()
pharmacy, challenge, window, deadline,
doctor licence, class, fill, cap,
commit(drug, quantity, salt)
```

Only the patient's secret can make these nullifiers, so two fills by one person look like fills by two people.

THE PRIVACY BOUNDARY

Who learns what

	PUBLIC LEDGER	PATIENT'S PHONE	PHARMACY
Who the patient is	**Never**	Holds the secret	**Never**
Medicine and quantity	A sealed commitment	Yes	At the counter, to hand it over
Expiry date	**Never (a 30-minute deadline only)**	Yes	No
Prescribing doctor	Licence number	Yes	Licence number
Drug class	Yes, for volume statistics	Yes	Yes
The patient's other prescriptions	**Never**	Yes	**Never**

Every value on the ledger passes through disclose() in the contract. docs/PRIVACY-BOUNDARY.md checks it line by line.

WHY MIDNIGHT

Shared rules over private inputs, *with nobody in the middle*

Shared, verifiable state

The cap has to hold across pharmacies that do not trust each other. Trees, nullifiers and fill records live on Midnight's public ledger.

Private witnesses

The patient's secret and the prescription stay in private state on the phone. Compact proves the rules over them and publishes only what `disclose()` marks.

Authorization by proof, not wallet

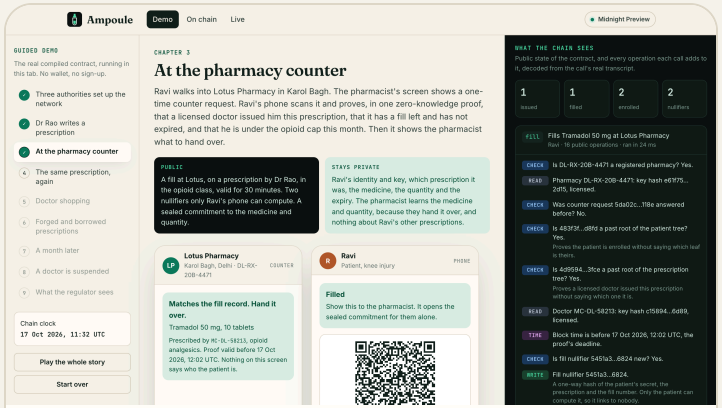
Doctors, authorities and patients prove a secret whose hash is on the ledger. Whoever pays the DUST fee learns nothing and links nothing.

Refusals cost nothing

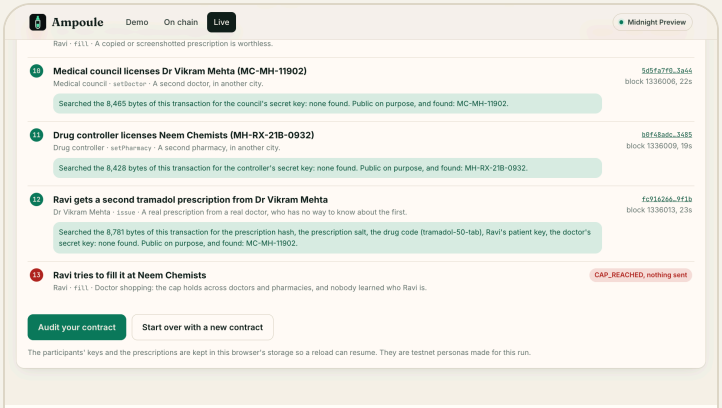
A fraudulent fill fails on the caller's device while the circuit runs. No proof, no transaction, no fee, no trace.

THE PRODUCT

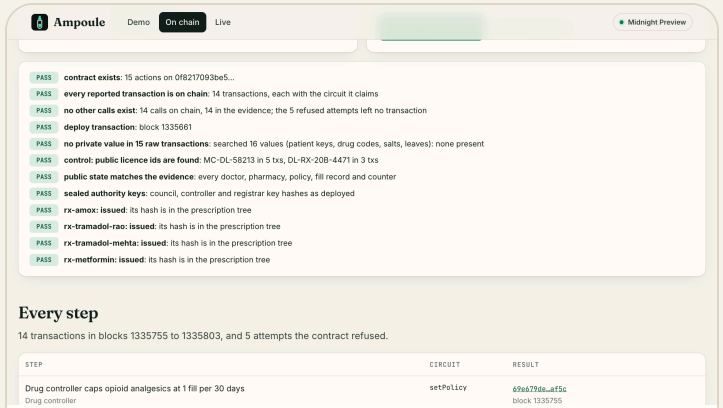
Three ways to see it work



Guided demo. Nine chapters on the compiled contract, in the browser, no wallet. A panel decodes every transcript and searches it for the private inputs.



Live with Lace. Deploy your own contract on Preview and run the story with real proofs. Each transaction's raw bytes are searched for secrets.



On-chain audit. Downloads every transaction of the Preview deployment and checks state, history and leaks in your browser.

BUILT AND TESTED

Evidence, not claims

6

circuits in about 300 lines of Compact, compiled with compactc 0.31.1

77

contract and SDK tests, including a model-based test of random histories and decoded-transcript privacy tests

15

transactions on Midnight Preview with real proofs, plus 5 refused attempts that left nothing on chain

0

private values found in the raw bytes of any transaction of the deployed contract

CI recompiles the contract and requires the committed code to match, then runs every test, the typecheck, the build and Playwright on desktop and phone.

One SDK drives the in-browser simulator, the CLI on Preview and the Lace live page. Each participant holds only their own secret.

A public audit anyone can run, from the CLI or a button on the site, needing no wallet and no key.

WHO IT IS FOR

Regulators buy it. *Pharmacies use it.*

Drug controllers and councils

License, cap and suspend with one transaction each. Get live public counts per doctor and pharmacy, and no register to guard.

Pharmacies and their software

One scan at the counter that covers every other pharmacy. The fill record replaces the paper register entry.

Doctors and patients

Issuing is one SDK call from e-prescription software. The patient carries a QR code, like the slip.

12 lakh+

chemists and distributors in India's national trade association (AIOCD)

79.71 cr

ABHA health IDs by July 2025: the registrar that enrolls one key per person

2018

tramadol notified under the NDPS Act; the controlled list keeps growing

Business model: a fee per controlled fill paid by the pharmacy or programme, a regulator dashboard, and SDK licences for software vendors. Fees are paid in DUST, so patients never handle tokens. Sources in docs/ADOPTION.md.

ROADMAP

From a working contract to a working counter

WAVE 2, DONE

- Compact contract, 6 circuits, 77 tests
- SDK, CLI and Preview deployment
- Guided demo, Lace live page, public audit
- Privacy boundary, threat model, adoption path

WAVE 3

- Patient wallet as an installable web app
- Counter page with camera scanning
- Pharmacy-paid fees, so no wallet links fills
- Patient key rotation, rounded deadlines

AFTER THE BUILDATHON

- Pilot: one hospital and its nearby pharmacies, one or two controlled classes
- Plug-ins for e-prescription and pharmacy billing software
- More classes, more states, one contract

TRY IT

Policing without the register.

Demo video	youtu.be/CUaj5al-Ff4
Guided demo	sanchay117.github.io/ampoule/#/demo
Live with Lace	sanchay117.github.io/ampoule/#/live
On-chain audit	sanchay117.github.io/ampoule/#/ledger
Code	github.com/Sanchay117/ampoule
For judges	JUDGES.md, a five-minute path

Apache-2.0. All people and pharmacies in the demo are fictional. Testnet only.